

Capstone AI Agent Specification

PROJECT: ResearchScout Agent | PHASE: Build (Core) | ESTIMATED: 10 Hours | AUTHOR: Abdul Sami

1. Job to be Done & Scope

Agent Name: *ResearchScout Agent*
Primary Job: Autonomous research assistant that monitors AI/ML literature (arXiv, PapersWithCode), extracts methodology frameworks, synthesizes papers into structured markdown briefs, and highlights implementation gotchas.
User & Usage: ML Engineer (Single user). Used daily for automated digests and on-demand for specific arXiv ID deep dives.
Scope (10 Hours): Setup arXiv parser (3h) → Build synthesis engine (3h) → Implement guardrails & evals (2h) → CLI & test runs (2h).

2. Tools, Data Sources & Access Plan

- **arXiv REST API:** Public REST endpoint for metadata/query fetching (Free, no authentication).
- **PyPDF2 / PDFPlumber:** Local Python PDF parsing library for document extraction.
- **Gemini 2.5/3 API:** LLM inference and structured function calling engine (Free tier API key).
- **Local Markdown Store:** File system storage (work/outputs/research_briefs/) for persistent reports.

3. Draft System Instructions (System Prompt)

You are ResearchScout, an expert AI research assistant. Your sole role is to ingest technical AI papers and produce rigorous, hallucination-free engineering briefs.

- Strict Operational Guidelines:
1. Ground all analysis strictly in provided paper text.
 2. Structure output into: (A) Key Innovation, (B) Architecture Breakdown, (C) Empirical Results, and (D) Implementation Gotchas.
 3. Never fabricate benchmark metrics or cite unverified claims.
 4. Explicitly declare missing or corrupted data sections.

4. Five Pre-Build Eval Cases

ID	Scenario	Input Trigger	Expected Output & Criteria
E1	Specific arXiv ID Lookup	analyze_paper('2312.00752')	Extracts Mamba SSM architecture details vs Transformer attention.
E2	Topic Synthesis	search_topic('LoRA efficiency')	Returns top 3 benchmarked papers with comparison matrix.
E3	Out-of-Domain Refusal	"Write a sales cold email"	Politely refuses and redirects to research capabilities.
E4	Corrupted PDF Fallback	Invalid PDF URL	Triggers graceful exception handler without crashing pipeline.
E5	Math Formula Extraction	Paper with heavy LaTeX math	Extracts core equations and translates into plain intuition.

5. Risks & Guardrails

- **Risks Identified:** Hallucinated metrics, execution of untrusted paper code, accidental document overwrites.
- **Must Confirm:** Must prompt user before overwriting existing report files or executing local scripts.
- **Must Never Do:** Never execute unverified code snippets found in papers; never alter system credentials.

6. Platform Choice & Justification

- **Chosen Platform:** Custom Python Scripted Agent (Python + Gemini SDK + LangChain Tools).
- **Alternative Evaluated:** Claude Cowork / Custom GPTs (Paid platforms).
- **Justification:** 100% free execution, native access to local markdown repository, full programmatic control over evaluation harnesses, zero subscription lock-in.